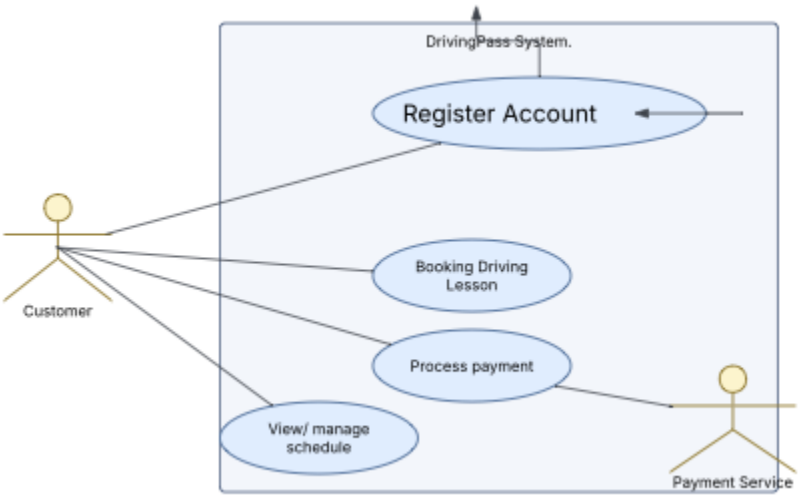
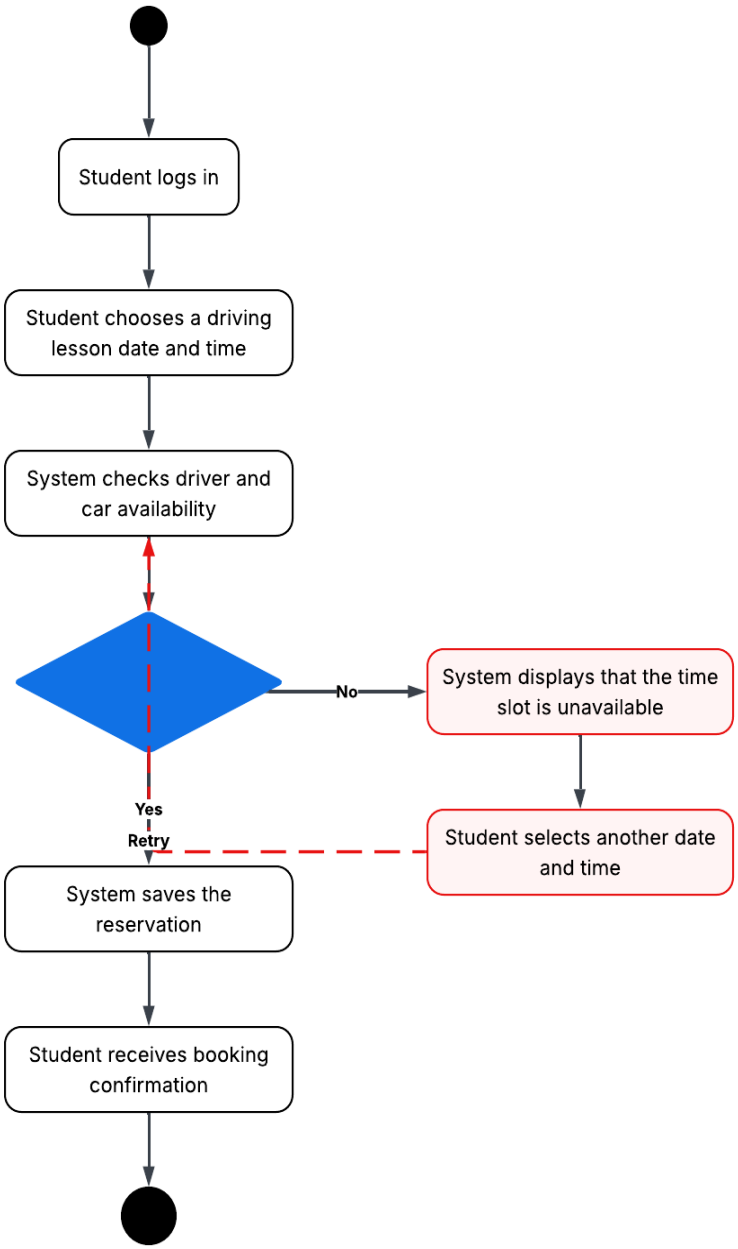
CS 255 System Design Document Template

UML Diagrams

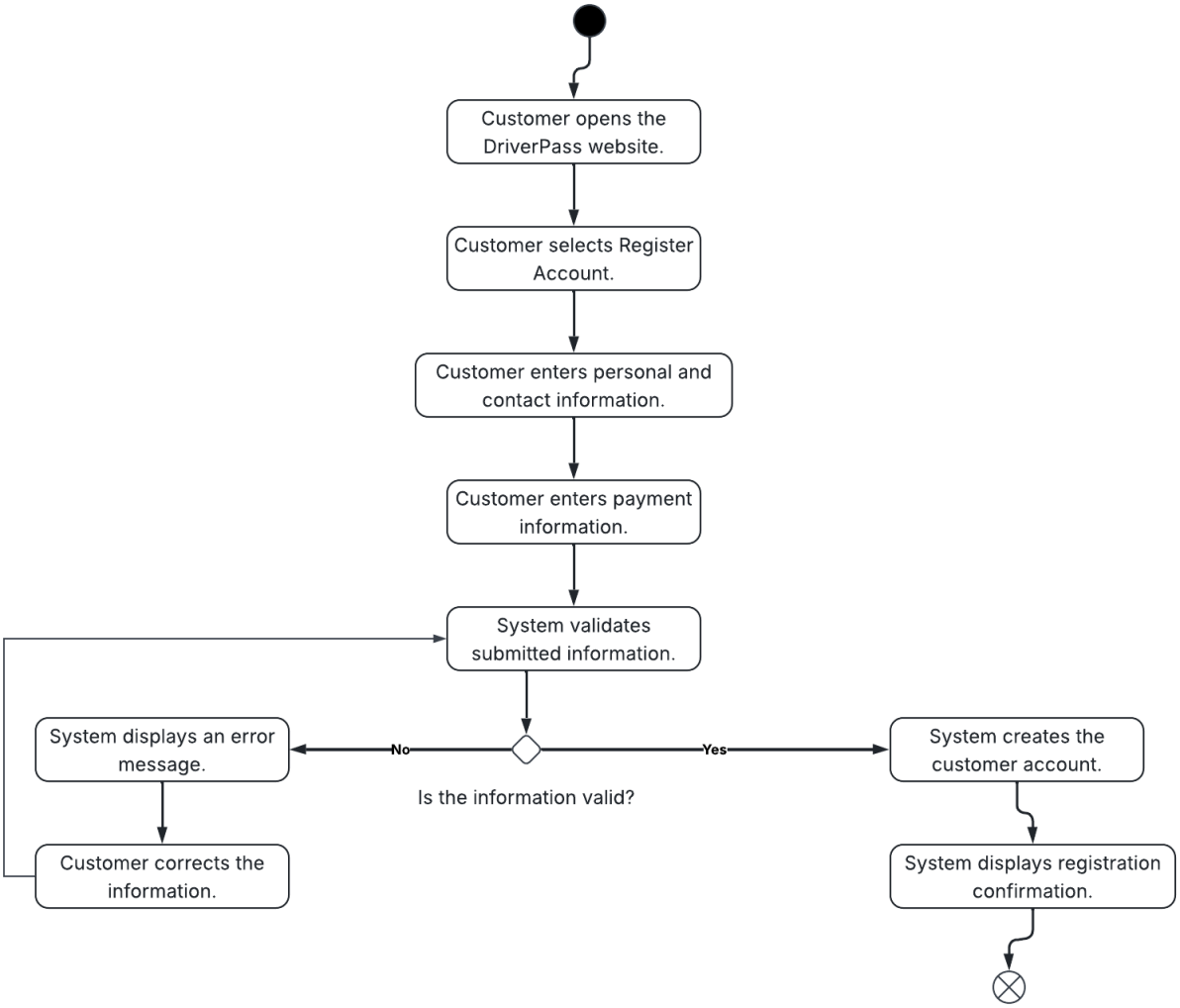
UML Use Case Diagram



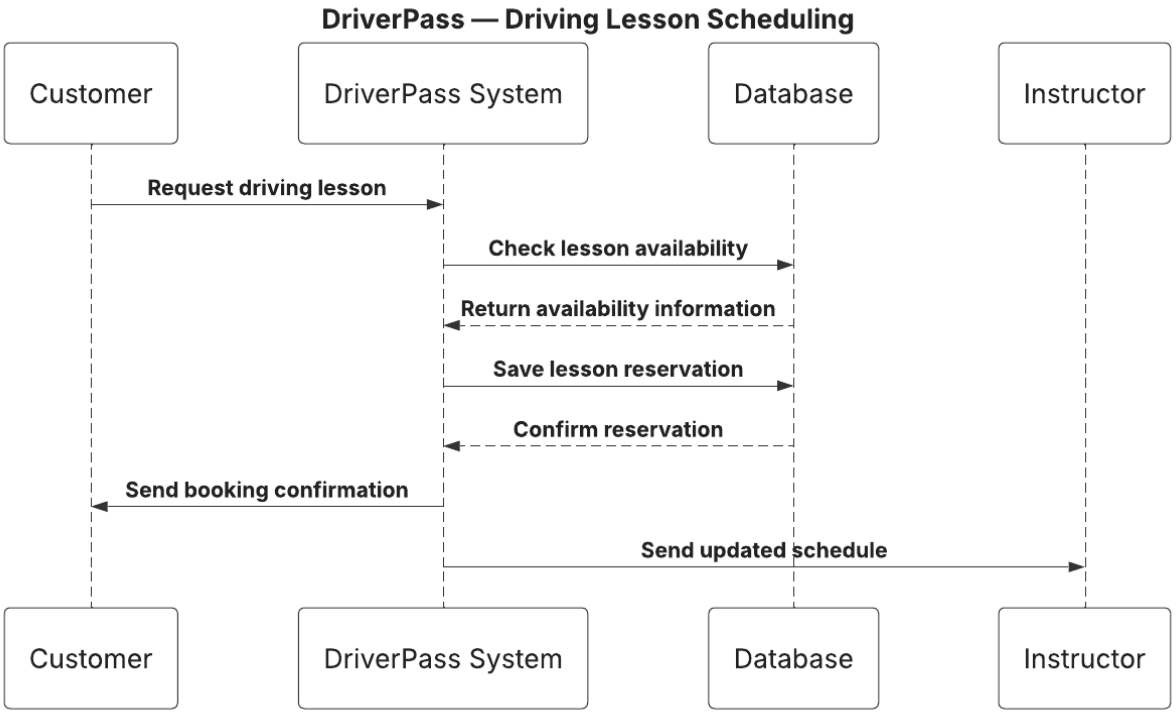
UML Activity Diagrams



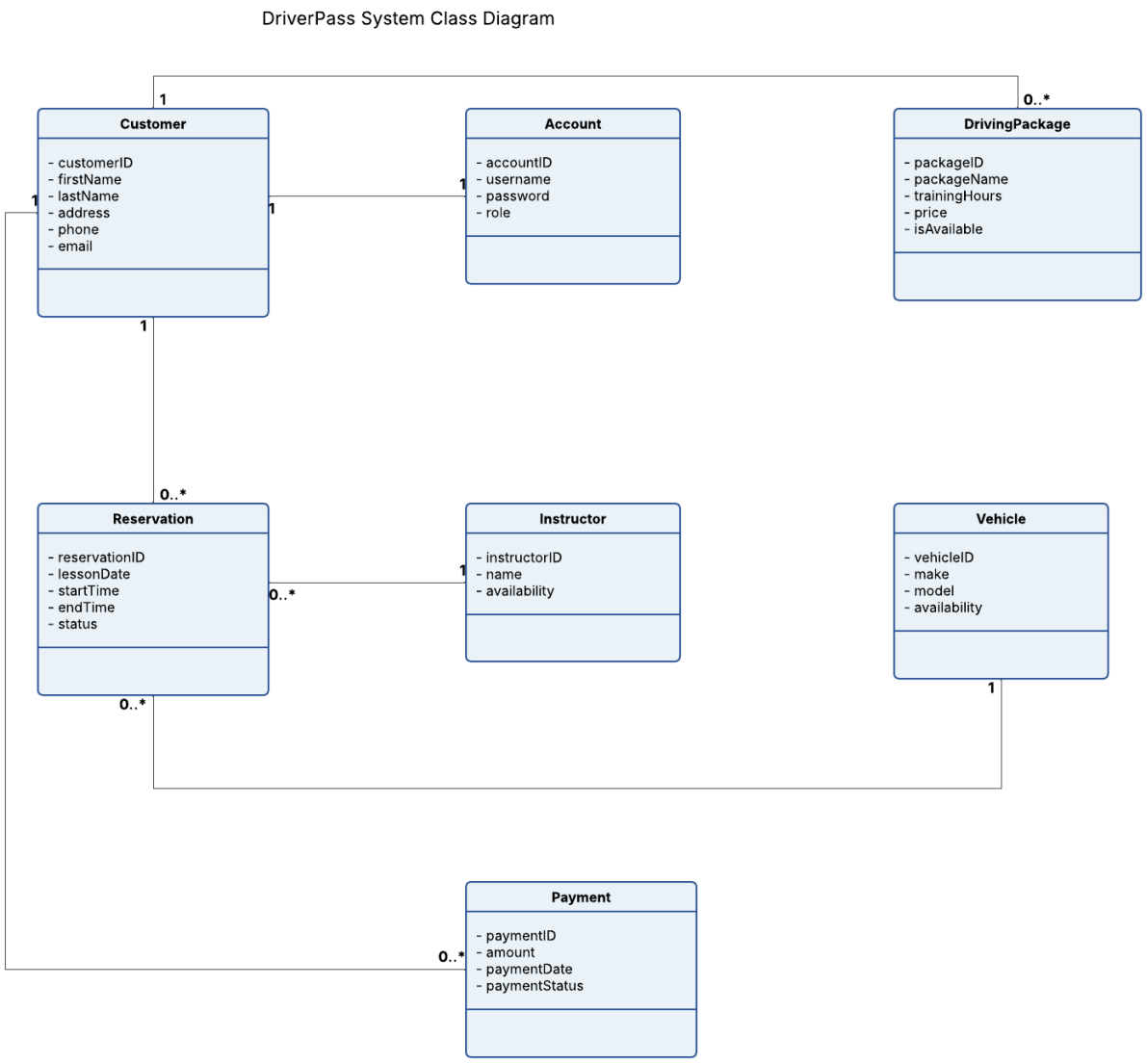
DriverPass — Register an Account



UML Sequence Diagram



UML Class Diagram



Technical Requirements:

The DriverPass system will be a website that customers and employees can use from a computer, tablet, or phone as long as they have internet access. The system will need cloud hosting, a web server, and a database to store customer accounts, driving packages, lesson schedules, payments, and student progress.

Customers should be able to create an account, purchase a driving package, schedule lessons, make payments, and check their progress online. Employees will need to manage lesson schedules and update student information. The system should also have different access levels so employees can only access the information they need. The IT officer will need access to manage accounts and passwords, while the owner should be able to view reports and see who made changes to reservations.

Security is important because the system will store personal and payment information. It should use HTTPS to protect information sent through the website, secure login features, and regular backups to prevent data from being lost. Payments should be handled through a secure payment service. The scheduling system also needs to check if a driver and car are available before confirming a lesson so that two people do not book the same time slot.

The system should work with common web browsers and include a way to export reports that can be opened in Excel. It should also allow updates to DMV rules, policies, and practice-test questions. Since the system will be hosted in the cloud, the company can maintain and update it without requiring customers to install anything on their devices.